MA 232 Course Project 2: How AI Detects Objects

A Linear Algebra Approach to Pattern Recognition

I. Objective & Background

Pattern recognition is an important task in image processing, computer vision, and data science. A simple image can be represented by a grid of colored cells, and the colors can be encoded numerically as a vector.

In this project, you will use concepts from linear algebra to construct a matrix that recognizes one specific pattern among all possible blue-and-white patterns on a 4×4 grid.

You will represent the target pattern as a vector $w \in \mathbb{R}^{16}$, construct a 16×16 matrix M, and use the quadratic form

$$Q(u) = u^T M u$$

to distinguish the target pattern from other patterns.

II. Submission Requirements

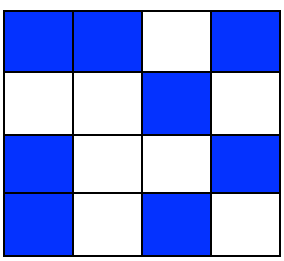
1) Your report must contain:
- The target grid and its vector representation.
- A clear explanation of the pattern-encoding procedure.
- Your 16×16 matrix and the reasoning behind its construction.
- A mathematical verification of the recognition condition.
- A table of test patterns and quadratic-form values.
- Your second matrix and investigation of matrix structure.
- Your reflection and conclusions.
- (You don't need to include code or spreadsheet formulas used.)

2) Your report must include mathematical explanations. A matrix and a collection of software-generated values without justification will not receive full credit.

3) You can use any computational software or tools for your calculations.

III. Project Tasks

Given the following pattern.



Use the following encoding:

- Blue cell = 1
- White cell = 0

Part A: Convert the pattern into a vector

1) Convert the given grid into a vector $w \in \mathbb{R}^{16}$.
2) How many blue cells and white cells does the target pattern contain?
3) Explain how an arbitrary blue-and-white 4×4 grid can be represented by a vector $u \in \{0, 1\}^{16}$.

Part B: Construct a matrix for pattern recognition

Your goal is to construct a 16×16 matrix M such that the quadratic form

$$Q(u) = u^T M u$$

recognizes the target pattern.

The matrix must satisfy the following conditions:

$$w^T M w = 0$$

and

$$u^T M u \neq 0$$

for every other nonzero binary vector ($u \in \{0, 1\}^{16}$ with $u \neq w$.

Complete the following tasks.

1) Explain what information is represented by the vector u.
2) Describe the role of the quadratic form $Q(u) = u^T M u$ in recognizing a pattern.
3) Construct a 16×16 matrix M that satisfies the two conditions.
4) Show that your matrix satisfies $w^T M w = 0$.
5) Explain why the quadratic form must be nonzero for every other nonzero binary vector.
6) State whether your matrix is symmetric and whether it is invertible. Justify your answers.

Part C: Verify the recognition rule

A matrix is not a successful solution merely because it recognizes the target pattern. You must verify that it does not incorrectly recognize another nonzero binary pattern.

1) Develop a systematic method to verify your matrix.

2) Verify the quadratic-form value for the target vector.

3) Test your matrix on several other binary patterns, including:

- A grid with only one blue cell.
- A grid with two blue cells.
- A grid with three blue cells in different positions.
- The grid with all 16 cells blue.

4) Explain how you can verify the condition for all $2^{16} - 1 = 65{,}535$ nonzero binary vectors.

5) Present a table of at least 10 test patterns, their binary vectors, and the corresponding quadratic-form values.

6) State how you know that your construction satisfies the requirement for all nonzero binary patterns, not merely the patterns in your sample table.

Part D: Explore different matrix constructions

1. **Construct a second matrix**
   - **a)** Find a second 16×16 matrix N, different from your original matrix N, that recognizes the same target pattern.
   - **b)** Verify that
   $$w^T N w = 0$$
   and
   $$u^T N u \neq 0$$
   for every other nonzero binary vector $u \neq w$.
   - **c)** Explain how the second matrix differs from the first and why both matrices work.

2. **Investigate matrix structure**
   Investigate whether the matrix must be diagonal to satisfy the recognition condition.
   - **a)** Can a matrix with nonzero off-diagonal entries satisfy the requirements?
   - **b)** Explain how the entries of a matrix contribute to the value of $u^T M u$.
   - **c)** Construct or describe a non-diagonal matrix that satisfies the recognition condition, if possible.

Part E: Reflection and applications

Write a reflection of approximately 300–500 words addressing the following questions:

- **a)** How does converting a visual grid into a vector make mathematical analysis possible?
- **b)** What was the most challenging part of constructing the matrix?
- **c)** How did you verify that the matrix recognizes only the target pattern?

d) Why is systematic verification important in pattern recognition?

e) How could a similar approach be used in image processing, data science, or another real-world application?

Part F: Group Performance Evaluation

This evaluation seeks your opinions about the level of contribution that you and the other team members on your team have made to your Course Project. Please answer the questions by circling the number that most closely matches your assessment. Please note that this evaluation is entirely confidential.

Scale

Excellent	Good	Fair	Poor	Not responsible at all
5	4	3	2	1

Your Name: _____

- Communication 5 4 3 2 1
- Leadership 5 4 3 2 1
- Quality of Work 5 4 3 2 1
- Timeliness for Meetings 5 4 3 2 1
- Completing work on time 5 4 3 2 1
- Provide feedback for other's work 5 4 3 2 1
- Respect for other's opinion 5 4 3 2 1
- The total (overall) work that all members in your group completed is 100%. What is the percentage of the part that you contributed? Write your answer as a percent.

Team Member 1's Name: _____

- Communication 5 4 3 2 1
- Leadership 5 4 3 2 1
- Quality of Work 5 4 3 2 1
- Timeliness for Meetings 5 4 3 2 1
- Completing work on time 5 4 3 2 1
- Provide feedback for other's work 5 4 3 2 1
- Respect for other's opinion 5 4 3 2 1

- The total (overall) work that all members in your group completed is 100%. What is the percentage of the part that this member contributed? Write your answer as a percent.

Team Member 2's Name: _____

- Communication 5 4 3 2 1
- Leadership 5 4 3 2 1
- Quality of Work 5 4 3 2 1
- Timeliness for Meetings 5 4 3 2 1
- Completing work on time 5 4 3 2 1
- Provide feedback for other's work 5 4 3 2 1
- Respect for other's opinion 5 4 3 2 1
- The total (overall) work that all members in your group completed is 100%. What is the percentage of the part that this member contributed? Write your answer as a percent.